KARNATAKA SCHOOL EXAMINATION & ASSESSMENT BOARD

2026 II PUC EXAM-2

SUBJECT: **34-CHEMISTRY** SCHEME OF EVALUATION MAX. MARKS: 70

	PART-A	
I.	**Select the correct option from the given choices:**	**15 × 1=15**
1)	The mixture which shows positive deviation from Raoult's Law is a) n-hexane and n-heptane b) Bromoethane and Chloroethane c) Ethanol and Acetone d) Chloroform and Acetone	
Ans:	**c) Ethanol and Acetone (or) Ethanol and Acetone (or) c)**	**1**
2)	Which of the following is homoleptic complex? a) $\left[Co(NH_3)_3Cl_3\right]$ b) $\left[Co(NH_3)_5Cl\right]Cl_2$ c) $\left[Co(NH_3)_4(H_2O_2)\right]Cl_3$ d) $\left[Co(NH_3)_6\right]Cl_3$	
Ans:	**d)** $\left[Co(NH_3)_6\right]Cl_3$ **(or)** $\left[Co(NH_3)_6\right]Cl_3$ **(or) d)**	**1**
3)	The difference between the electrode potentials of two electrodes, when no current is drawn through the cell is called a) Cell potential b) Cell emf c) Potential difference d) Standard electrode potential	
Ans:	**b) Cell emf (or) Cell emf (or) b)**	**1**
4)	The unit of rate equation for gaseous reaction is expressed in a) $atm\,s^{-1}$ b) $L\,atm\,s^{-1}$ c) $atm\,mol^{-1}\,s^{-1}$ d) $mol^{-1}\,s^{-1}$	
Ans:	**a)** $atm\,s^{-1}$ **(or)** $atm\,s^{-1}$ **(or) a)**	**1**
5)	The chemical formula of Hinsberg's reagent is a) C_6H_6SOCl b) $C_6H_5SOCl_2$ c) $C_6H_5SO_2Cl$ d) $C_6H_5SO_3H$	
Ans:	**c)** $C_6H_5SO_2Cl$ **(or)** $C_6H_5SO_2Cl$ **(or) c)**	**1**
6)	Cumene hydroperoxide on hydrolysis of dilute acids gives a) phenol and Oxygen b) Phenol and Hydrogen c) Hydrogen and Oxygen d) Phenol and Acetone	
Ans:	**d) Phenol and Acetone (or) Phenol and Acetone (or) d)**	**1**
7)	The best reagent useful for separation and purification of aldehyde from ketone is a) Tollens' reagent b) Sodium hydrogen sulphite c) Sodium sulphate d) 2,4-DNP reagent	
Ans:	**b) Sodium hydrogen sulphite (or) Sodium hydrogen sulphite (or) b)**	**1**
8)	The function of enzymes in the living system is to a) catalyse biochemical reactions b) provide energy c) provide immunity d) transport oxygen	
Ans:	**a) catalyse biochemical reactions (or) catalyse biochemical reactions (or) a)**	**1**
9)	**Statement I:** NO_2^- and SCN^- ions are ambidentate ligands. **Statement II:** NO_2^- and SCN^- ions can coordinate only through N to a central metal atom or ion. a) Both Statement I and Statement II are correct. b) Statement I is correct but Statement II is incorrect. c) Both Statement I and Statement II are incorrect. d) Statement I is incorrect but Statement II is correct.	
Ans:	**b) Statement I is correct but Statement II is incorrect. (or) (or) Statement I is correct but Statement II is incorrect. b)**	**1**

10) Match the following given in List I with List II

List I	List II
i) Vitamin A	A) Rickets
ii) Vitamin D	B) Pernicious anaemia
iii) Vitamin B	C) Increased blood clotting time
iv) Vitamin K	D) Xerophthalmia

Choose the correct option.

a) i-D, ii-A, iii-B, iv-C b) i-B, ii-D, iii-A, iv-C

c) i-D, ii-A, iii-C, iv-B d) i-A, ii-D, iii-B, iv-C

Ans: **a) i-D, ii-A, iii-B, iv-C (or) i-D, ii-A, iii-B, iv-C (or) a)** **1**

11) Acetylation of salicylic acid produces a compound which has anti-inflammatory and antipyretic property. The compound is

a) Bakelite b) Acetic anhydride c) Formalin d) Aspirin

Ans: **d) Aspirin (or) Aspirin (or) d)** **1**

12) A compound 'X' with molecular formula C_2H_4 reacts with HBr to give compound 'Y', which further reacts with NaI in the presence of dry acetone and gives compound 'Z'. The compound 'Z' is

a) C_2H_6 b) C_2H_5Br c) C_2H_5I d) C_2H_5F

Ans: **c) C_2H_5I (or) C_2H_5I (or) c)** **1**

13) Transition metal 'A' forms metal oxide with oxygen which is acidic and covalent. Its oxoanion oxidises iodide ion to iodate ion in neutral medium. The metal 'A' is

a) Mn b) Sc c) Ag d) Cr

Ans: **a) Mn (or) Mn (or) a)** **1**

14) The major product obtained and the type of reaction involved when $(CH_3)_3CBr$ reacts with CH_3ONa are

a) t-butyl methyl ether and nucleophilic substitution reaction
b) t-butyl methyl ether and elimination reaction
c) 2-methylpropene and nucleophilic substitution reaction
d) 2-methylpropene and elimination reaction

Ans: **d) 2-methylpropene and elimination reaction**
(or) 2-methylpropene and elimination reaction (or) d) **1**

15) In the three beakers labelled as A, B and C, 200 ml of water, 200 ml of 0.1M solution of sucrose in water and 200 ml of 0.01M solution of sucrose in water are taken respectively and kept at same temperature. Then

a) Vapour pressure in all three beakers are same
b) Vapour pressure of beaker 'C' is lowest
c) Vapour pressure of beaker 'B' is highest
d) Vapour pressure of beaker 'B' is lower than that of beaker 'C' and vapour pressure of beaker 'C' is lower than that of beaker 'A'

Ans: **d) Vapour pressure of beaker 'B' is lower than that of beaker 'C' and vapour pressure of beaker 'C' is lower than that of beaker 'A' (or)**
(or) Vapour pressure of beaker 'B' is lower than that of beaker 'C' and vapour pressure of beaker 'C' is lower than that of beaker 'A' (or) d) **1**

II. **Fill in the blanks by choosing the appropriate word from those given in the brackets: [primary, ethoxyethane, mass by volume percentage, chloroform, cerium, tertiary]** **5×1=5**

16) The unit of concentration of solutions commonly used in medicine and pharmacy is _____.

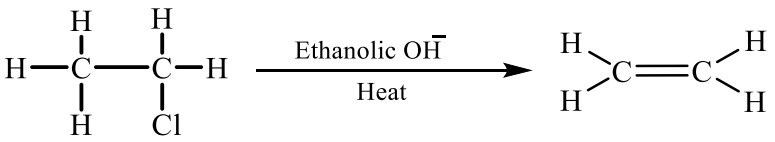
Ans: **mass by volume percentage** **1**

17)	A member of lanthanoid series which is well known to exhibit +4 oxidation state is _____.	
Ans:	**cerium**	**1**
18)	_____ is the symmetrical ether.	
Ans:	**ethoxyethane**	**1**
19)	The polyhalogen compound used as anaesthetic during surgery was _____.	
Ans:	**chloroform**	**1**
20)	Gabriel synthesis is used for the preparation of _____ amines.	
Ans:	**primary**	**1**
	PART - B	
III.	**Answer any three of the following. Each question carries 2 marks.**	**3 × 2 = 6**
21)	**Write two differences between order and molecularity of a reaction.**	

Ans:

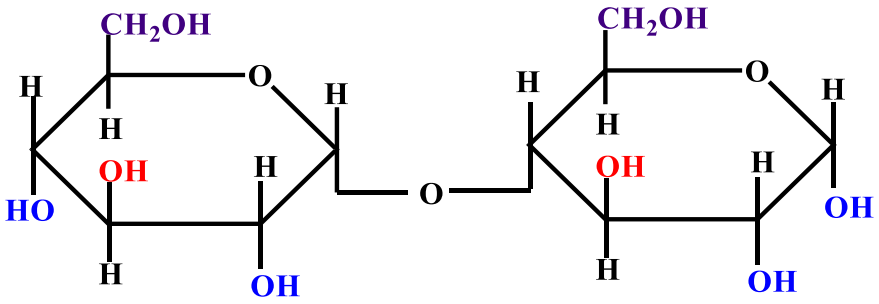
Order of a reaction	Molecularity of a reaction
The sum of the powers concentration terms of reactants in the experimentally determined rate equation.	It is the number of reacting species involved in elementary or simplest reaction.
It is experimentally determined.	It is theoretically calculated.
It can be zero and even a fraction.	It cannot be zero or a non-integer.
Order is applicable to elementary as well as complex reactions	Molecularity is applicable only for elementary reactions.
For complex reaction, order is given by the slowest step.	Molecularity of the slowest step is same as the order of the overall reaction.

(Any two differences, 1 mark each) **2**

22)	**What is Lanthanoid contraction? Mention one consequence of it.**	
Ans:	The overall decrease in atomic radii and ionic radii from lanthanum to lutetium (across lanthanoids) is called Lanthanoid contraction.	**1**
	1. The separation of lanthanoids in pure state becomes difficult. **(or)** Difficulty in separation of lanthanoids due to similar chemical properties.	
	2. The atomic radii of 3rd row transition series elements are almost similar to that of 2nd row transition series elements. **(or)** The identical radii of Zirconium (Zr) and Hafnium (Hf).	**1**
	3. The covalent character increases.	
	4. Basicity decreases. **(Any one correct consequences: 1 mark)**	
23)	**Explain dehydrohalogenation reaction of haloalkanes with an example.**	
Ans:	Dehydrohalogenation is an elimination reaction that removes a hydrogen atom and a halogen atom from adjacent carbons in a haloalkane in presence of ethanolic base to form an alkene.	**1**
		
	(Any suitable example; Statement 1 mark and equation 1 mark (or) self-explanatory equation 2 mark)	**1**
24)	**What is Lucas reagent? Which class of alcohols does produce turbidity immediately with it?**	
Ans:	The mixture of Conc. HCl and Anhyd. $ZnCl_2$ **(or)** Conc. HCl and $ZnCl_2$ is called Lucas reagent.	**1**
	Tertiary alcohols	**1**

25) **Write the Haworth structure of Maltose.**

Ans:



2

PART - C

IV. **Answer any three of the following. Each question carries 3 marks.** **3 × 3 = 9**

26) **Write the balanced chemical equations involved in the manufacture of potassium dichromate from chromite ore.**

Ans:

Step-1: $\mathbf{4}\ FeCr_2O_4 + \mathbf{8}\ Na_2CO_3 + \mathbf{7}\ O_2 \rightarrow \mathbf{8}\ Na_2CrO_4 + \mathbf{2}\ Fe_2O_3 + \mathbf{8}\ CO_2$ **1**

Step-2: $\mathbf{2}\ Na_2CrO_4 + \mathbf{2}\ H^+ \rightarrow Na_2Cr_2O_7 + \mathbf{2}\ Na^+ + H_2O$

(or)

$\mathbf{2}\ Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$ **1**

Step-3: $Na_2Cr_2O_7 + \mathbf{2}\ KCl \rightarrow K_2Cr_2O_7 + \mathbf{2}\ NaCl$ **1**

[Note: $FeCr_2O_4$ can also be written as $FeO.Cr_2O_3$]

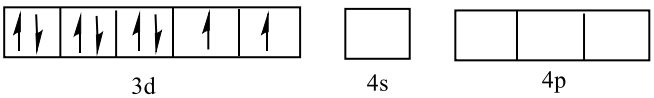
27) **a) Transition metals show good catalytic property. Give any two reasons.**
b) What are interstitial compounds?

Ans: a)
1. Due to small atomic size.
2. Due to presence of large surface area.
3. They exhibit variable oxidation states. **(Any two; 1 mark each)** **2**

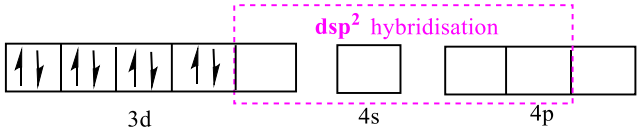
b) Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. **1**

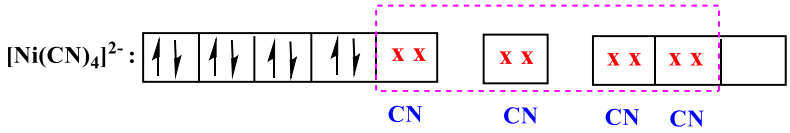
28) **On the basis of Valence Bond Theory, explain the hybridisation, geometry and magnetic property of complex ion $\left[Ni(CN)_4\right]^{2-}$. [Given : Atomic number of Ni = 28]**

Electronic configuration of Ni^{2+}: $[Ar]\ 3d^8\ 4s^0$



Since CN^- is strong field ligand therefore electrons pairing takes place



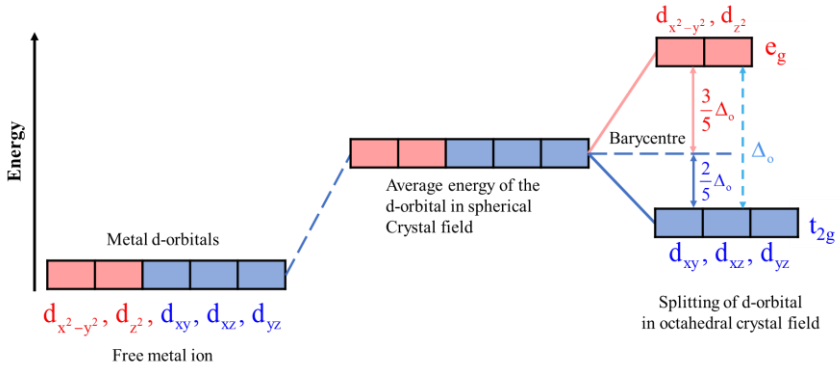


Hybridisation: dsp^2. **Magnetic property:** Diamagnetic. **Geometry:** Square planer.

Geometry: one-mark, Magnetic property: one mark; Explanation: one mark

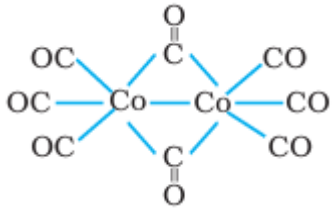
29) **(a) Draw the energy level diagram for the splitting of d-orbital in octahedral complex.**
(b) Write the structure of octacarbonyldicobalt (0).

Ans: a)



2

b)



1

30) **For the complex $\left[Fe(en)_2Cl_2\right]Cl$ (Atomic number of Fe = 26)**

(a) What is the oxidation state of metal ion?

(b) Mention the coordination number.

(c) How many moles of $AgNO_3$ is required to react with one mole of the complex?

Ans:

(a) +3 **1**

(b) 6 (or) six **1**

(c) 1 (or) one **1**

V **Answer any two of the folowing. Each question carries 3 marks:** **2 × 3 = 6**

31) **Write any three differences between ideal and non-ideal solutions.**

Ans:

	Ideal-solutions	Non-ideal solutions
1	They Obey Raoult's law at all temperature and pressure.	They do not obey Raoult's law.
2	$\Delta H_{mix} = 0$.	$\Delta H_{mix} \neq 0$
3	$\Delta V_{mix} = 0$.	$\Delta V_{mix} \neq 0$.
4	$P_{total} = P_A^0\chi_A + P_B^0\chi_B$	$P_{total} \neq P_A^0\chi_A + P_B^0\chi_B$
5	Intermolecular attractive forces between A-A, B-B and A-B are same.	Intermolecular attractive forces between A-A, B-B and A-B are not same.
	(Any three differences; for each 1 mark)	

3

32) **Derive an integrated rate equation for the rate constant of a zero-order reaction.**

Ans:

Consider a reaction: $R \rightarrow P$

$$Rate = \frac{-d[R]}{dt} = k[R]^0$$ **1**

$$Rate = \frac{-d[R]}{dt} = k \times 1$$

$d[R] = -k\ dt$

Integrate on both sides

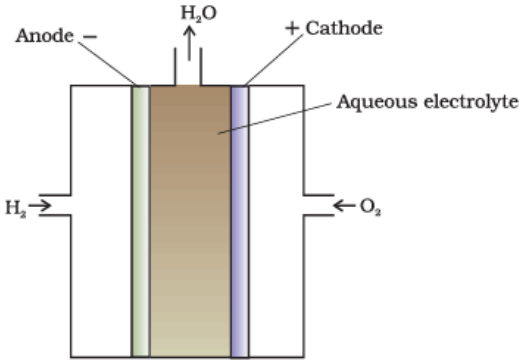
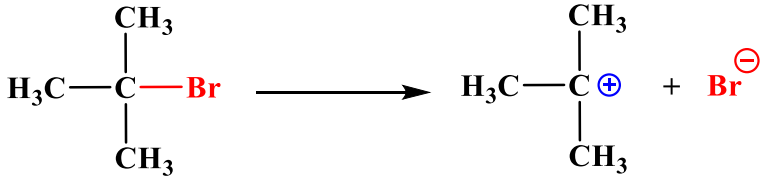
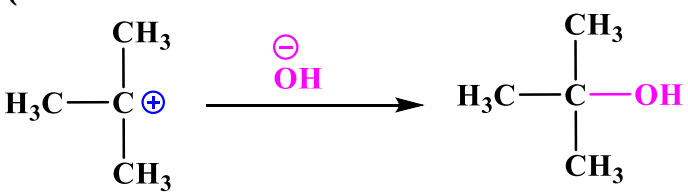
$[R] = -k\,t + I$ ---------- (1) **1**

At, $t = 0$, then $[R] = [R]_0$

Where, $[R]_0$ is the initial concentration of reactants; Substituting in equation (1)

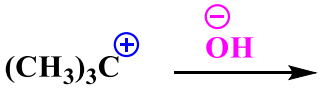
$[R]_0 = -k \times 0 + I$

$[R]_0 = I$

	Substituting the value in I in the equation (1) $[R] = -k\,t + [R]_0$ $-k\,t = [R] - [R]_0$ (or) $k = \frac{[R]_0 - [R]}{t}$ **(Any correct alternate derivations marks should be awarded)**	**1**
33)	**What are fuel cells? Draw a neat labelled diagram and write an overall cell reaction of H_2-O_2 fuel cell.**	
Ans:	Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, methane, methanol, etc. directly into electrical energy are called fuel cells.	**1**
		**1**
	Overall reaction: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	**1**
34)	**The Daniell cell converts the chemical energy released during the redox reaction,** $Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ **to electrical energy and has an electrical potential equal to 1.1V, when concentration of Zn^{2+} and Cu^{2+} ions is unity. When an external opposite potential of greater than 1.1V is applied to the cell, then** **i) Predict the direction of flow of current** **ii) Specify whether zinc is deposited or dissolved at the respective electrode.** **iii) What type of cell is it?**	
Ans:	i) Zn to Cu	**1**
	ii) Zn is deposited	**1**
	iii) Electrolytic cell	**1**
	PART - D	
VI.	**Answer any four of the following. Each question carries 5 marks.**	**4 × 5 = 20**
35) a) **b)** **c)**	**Write the steps involved in S_N1 mechanism for the conversion of 2-Bromo-2-methylpropane into 2-Methylpropan-2-ol.** **Explain Fittig reaction with an example.** **Define chirality.**	**(2+2+1)**
Ans: a)		**1**
	(Marks should also be allotted for a mechanism with a reversible arrow for the above step)	
		**1**
	(or)	**(or)**

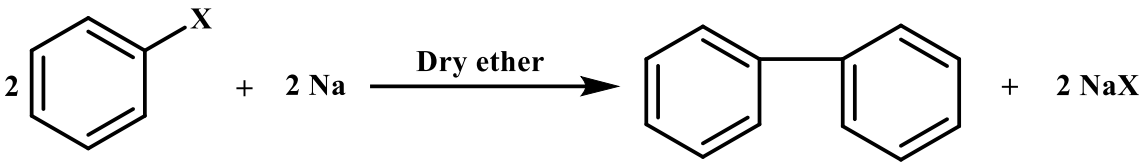


(Marks should also be allotted for a mechanism with a reversible arrow for the above step) **1**



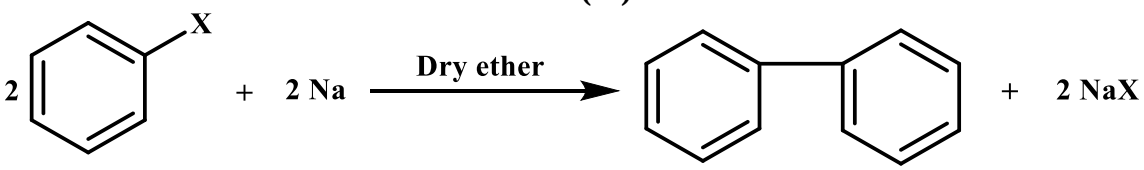
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b) When aryl halides react with sodium in presence of dry ether gives diphenyl. This reaction is called Fittig reaction. **1**



1

(or) **(or)**



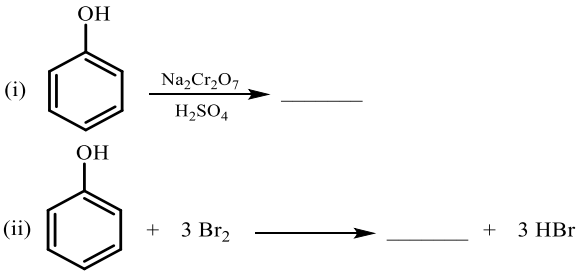
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Aryl halide **Diphenyl**

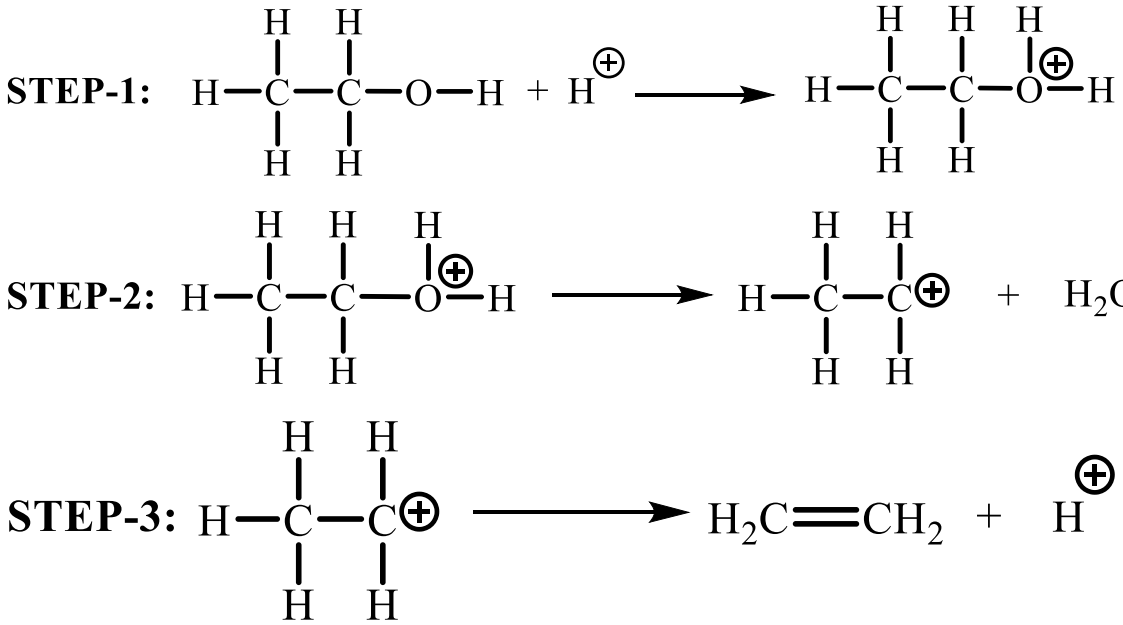
c) The objects which are non-super imposable on their mirror image are said to be chiral and this property is known as chirality. **1**

36) a) Write the steps involved in the mechanism of acid catalysed dehydration of ethanol to ethene.

b) Complete the following reactions; **(3+2)**



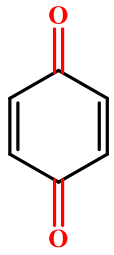
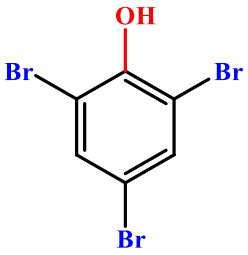
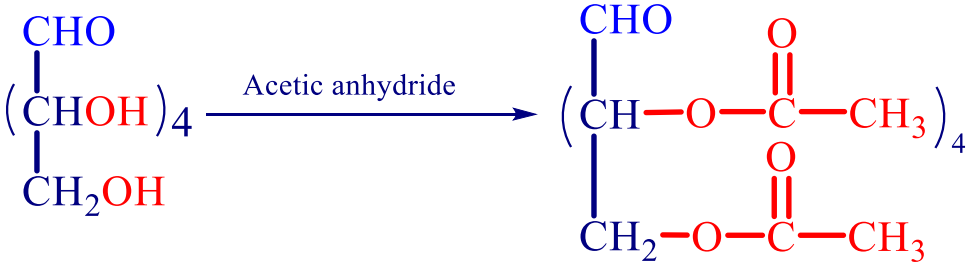
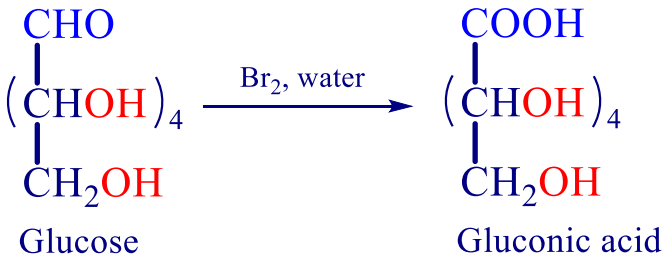
Ans: a)



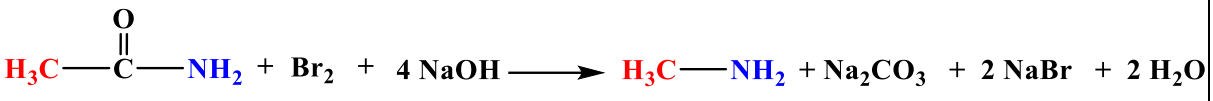
1 **1** **1**

(In this mechanism, charges are strictly expected where it is necessary and marks should also be allotted for a mechanism with a reversible arrow)

(or) **(or)**

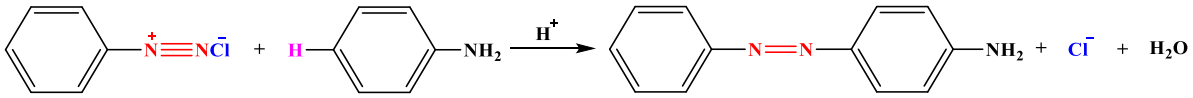
	$CH_3CH_2—OH + H^{\oplus} \longrightarrow CH_3CH_2—\overset{\oplus}{O}H_2$ $CH_3CH_2—\overset{\oplus}{O}H_2 \longrightarrow CH_3\overset{\oplus}{C}H_2 + H_2O$ $CH_3\overset{\oplus}{C}H_2 \longrightarrow H_2C{=}CH_2 + H^{\oplus}$ **(In this mechanism, charges are strictly expected where it is necessary and marks should also be allotted for a mechanism with a reversible arrow)**	**1** **1** **1**
b)	(i)  (ii) 	**1+1**
37) a)	**Write the reactions to show that glucose contains** **i) Five hydroxyl groups** **ii) An aldehyde group**	**(2+2+1)**
b)	**What are essential amino acids? Give an example.**	
c)	**Name the nitrogenous base present in RNA but not in DNA.**	
Ans: a)	i) 	**1**
	ii) 	**1**
b)	The amino acids, which cannot be synthesised in the body and must be obtained through diet, are known as essential amino acids. Example: Valine, Arginine, Lysine and etc.,	**1** **1**
c)	Uracil **(or)** (U)	**1**
38) a)	**How is methanamine prepared by Hoffmann's bromamide degradation reaction? Write the equation.**	**(2+2+1)**
b)	**How do you convert benzenediazonium chloride to p-aminoazobenzene? Give the equation.**	
c)	**Write the IUPAC name of $(CH_3)_2N-C_2H_5$.**	

Ans: a) When acetamide reacts with with bromine in an aqueous or ethanolic solution of sodium hydroxide gives ethanamine. **1**



1

b) When benzene diazonium chloride reacts with aniline yields p-aminoazobenzene. **1**



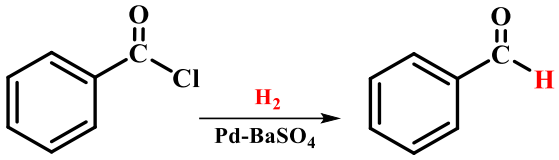
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(Statement 1 mark and equation 1 mark (or) self-explanatory equation 2 mark)

c) N.N-Dimethylethanamine **1**

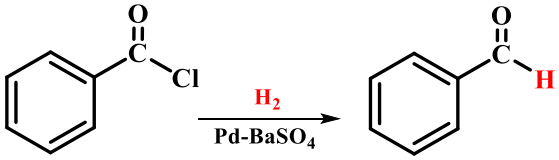
39) a) How is benzoyl chloride converted to benzaldehyde? Write the equation and name the reaction.
b) Explain Aldol condensation reaction with an example. **(3+2)**

Ans: a) Hydrogenation of benzoyl chloride over palladium on barium sulphate gives benzaldehyde. **1**



1

(or) **(or)**

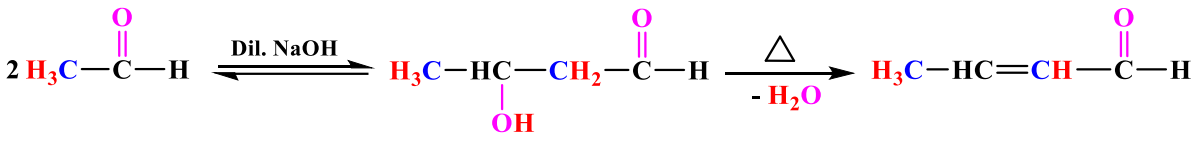


2

Benzoyl chloride **Benzaldehyde**

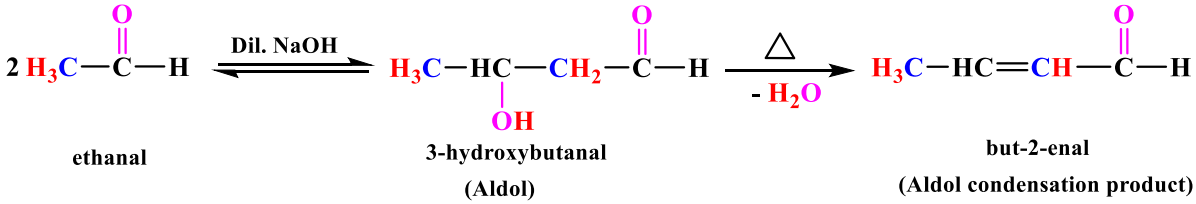
Rosenmund reduction **1**

b) The aldol or ketol on heated undergo dehydration to give α,β-unsaturated aldehyde or ketone. This reaction is called Aldol condensation. **1**



1

(or) **(or)**

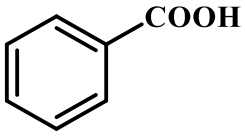
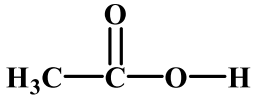
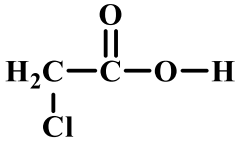
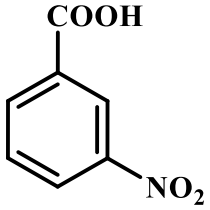


2

(Any suitable example; Statement 1 mark and equation 1 mark (or) self-explanatory equation 2 mark)

40) On hydrolysis of benzoic ethanoic anhydride gives two compounds A and B. Compound B on treatment with chlorine in the presence of red phosphorus followed by hydrolysis forms compound C. Compound A reacts with a mixture of concentrated nitric acid and concentrated sulphuric acid to give compound D. Identify A, B, C and D. Among A and B, which has higher pKa value?

Ans: a)

'A'	**C_6H_5COOH (or)**  **(or) Benzoic acid**	**1**
'B'	**CH_3COOH (or)**  **(or) Acetic acid**	**1**
'C'	**$CH_2ClCOOH$ (or)**  **(or) 2-chloroacetic acid**	**1**
'D'	 **(or) 3-nitrobenzoic acid (or) m-nitrobenzoic acid**	**1**
	'B' has higher pKa value	**1**

PART – E (PROBLEMS)

VII. **Answer any three of the following. Each question carries 3 marks.** **3 × 3 = 9**

41) **2.00 g of a non-electrolyte solute dissolved in 100 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant for benzene is 5.12 K kg mol^{-1}. Find the molar mass of the solute.**

Ans:

$$M_2 = \frac{1000 \times w_2 \times K_f}{\Delta T_f \times w_1} \quad \text{(or)} \quad M_B = \frac{1000 \times w_B \times K_f}{\Delta T_f \times w_A} \qquad \textbf{1}$$

$$M_2 = \frac{1000 \times 2 \times 5.12}{0.4 \times 100} \qquad \textbf{1}$$

$$M_2 = 256\,g\,mol^{-1} \qquad \textbf{1}$$

(Strictly expecting correct unit with final answer)

42) **The cell in which the reaction occurs**

$2Fe^{3+}_{(aq)} + 2I^-_{(aq)} \longrightarrow 2Fe^{2+}_{(aq)} + I_{2(s)}$; $E^o_{cell} = 0.230\,V$ at $298\,K$.

Calculate the value of log K_c (K_c = equilibrium constant) of the cell reaction.

Ans:

$$E^o_{Cell} = \frac{0.0591}{n} \times \log K_c \text{ (or) } \log K_c = \frac{E^o_{Cell} \times n}{0.0591} \quad [OR] \quad E^o_{Cell} = \frac{0.059}{n} \times \log K_c \text{ (or) } \log K_c = \frac{E^o_{Cell} \times n}{0.059} \qquad \textbf{1}$$

$$\log K_c = \frac{0.230 \times 2}{0.0591} \quad [OR] \quad \log K_c = \frac{0.230 \times 2}{0.059} \qquad \textbf{1}$$

$\log K_c = 7.7834$ (or) $\log K_c = 7.78$ [OR] $\log K_c = 7.7966$ (or) $\log K_c = 7.79$ **1**

43) **The rate constants of a reaction at 500 K and 700 K are 0.04 s^{-1} and 0.08 s^{-1} respectively. Calculate the value of E_a for the reaction. (Given: R = 8.314 $JK^{-1}mol^{-1}$)**

Ans:

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right) \qquad \textbf{1}$$

$$\log\left(\frac{0.08}{0.04}\right) = \frac{E_a}{2.303 \times 8.314}\left(\frac{700 - 500}{500 \times 700}\right) \qquad \textbf{1}$$

$E_a = 10085.75\,J\,mol^{-1}$ **(or)** $E_a = 10.08\,kJ\,mol^{-1}$ **1**

(Strictly expecting correct unit with final answer)

44)	**A first order reaction has half-life of 23 min. How much time will be needed for 30% completion of the reaction?**	
Ans:	$k = \frac{0.693}{t_{½}} = \frac{0.693}{23\,min} = 0.0301\,min^{-1}$ $t_{30\%} = \frac{1}{k}\ln\frac{[R]_0}{[R]}$ (or) $t_{30\%} = \frac{2.303}{k}\log\frac{[R]_0}{[R]}$ $t_{30\%} = \frac{1}{0.0301\,min^{-1}}\ln\left(\frac{100}{100-30}\right)$ (or) $t_{30\%} = \frac{2.303}{0.0301\,min^{-1}}\log\left(\frac{100}{100-30}\right)$ $t_{30\%} = 11.84\,min$ (or) $t_{30\%} = 710.4\ s$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**
45)	**H_2S, a toxic gas with rotten egg like smell, is used for the qualitative analysis. If the solubility of H_2S in water at STP is 0.195 m, calculate Henry's Law constant. (Given: p_{H_2S} at STP is 0.987 bar)**	
Ans:	$n_{H_2O} = \frac{1000\,g}{18} = 55.55\,mol$ $\chi_{H_2S} = \frac{n_{H_2S}}{n_{H_2S} + n_{H_2O}} = \frac{0.195}{0.195+55.55} = 0.00349$ $p = K_H\chi$ (or) $K_H = \frac{p}{\chi}$ $K_H = \frac{0.987}{0.00349}$ $K_H = 282.8\,bar$ **(Strictly expecting correct unit with final answer)**	**1** **1** **1**
46)	**The molar conductivity of 0.025 mol L^{-1} methanoic acid is 46.1 S cm^2 mol^{-1}. Calculate its degree of dissociation.** **Given:** $\Lambda^o_{(H^+)} = 349.6\,S\,cm^2mol^{-1}$ and $\Lambda^o_{(HCOO^-)} = 54.6\,S\,cm^2mol^{-1}$.	
Ans:	$\Lambda^o_{m(HCOOH)} = \lambda^o_{m(H^+)} + \lambda^o_{m(HCOO^-)}$ $\Lambda^o_{m(HCOOH)} = 349.6 + 54.6 = 404.2\,S\,cm^2mol^{-1}$ $\alpha = \frac{\Lambda_m}{\Lambda^o_m} = \frac{46.1}{404.2} = 0.114$	**1** **1** **1**

*********END*********